

Tutorial Problem Set V

MATA30 – TUT0003

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In the first part of the tutorial, we covered continuity, limits at infinity, asymptotes and the intermediate value theorem. The second half was spent on Quiz 2.

1 Basic problems

Problem 1. Compute the following limits.

(a) $\lim_{x \rightarrow \infty} \frac{\sin(x)}{x}$

(b) $\lim_{x \rightarrow -\infty} \frac{x-1}{\sqrt{2x^2+1}}$

(c) $\lim_{x \rightarrow \infty} \left(\sqrt{x + \sqrt{x}} - \sqrt{x} \right)$

Problem 2. Determine, with proof, the discontinuities of the following functions and the type of discontinuity.

(a) $f(x) = \begin{cases} e^{-\frac{1}{x^2}} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

(b) $f(x) = \begin{cases} \log|x-1| & \text{if } x \neq 1 \\ 0 & \text{if } x = 1 \end{cases}$

(c) $f(x) = \begin{cases} x^2 + x + 1 & \text{if } x \leq 2 \\ 11 - x^2 & \text{else} \end{cases}$

Problem 3. Compute the asymptotes of the function $f(x) = \frac{x^2}{x^2-x-2}$. Sketch the graph of f .

Problem 4. Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$ be a polynomial such that $a_0 < 0$ and $a_n + \cdots + a_0 > 0$.

- (a) Show that f has at least one zero on the interval $[0, 1]$.
- (b) Show that for every $k \geq 1$, there exists a polynomial f with the given properties, such that f has exactly k distinct zeroes on $[0, 1]$. *Hint:* You might want to use different constructions based on the parity of k .

2 Challenging problems

Problem 5. Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function. Prove that there exists $x \in [0, 1]$ such that $f(x) = x$.

Problem 6. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be continuous functions such that

$$\lim_{x \rightarrow \infty} (f(x) - g(x)) = 0.$$

- (a) Does it follow that $f = g$?
- (b) What if f and g are periodic with the same period?
- (c) What if f and g are periodic with different but rational periods?
- (d) What if f and g are periodic with different and possibly irrational periods?